A Project report Entitled

"WEED DETECTION AND DESTRUCTION MACHINE"

Submitted To,

Savitribai Phule Pune University (SPPU)

For the partial fulfillment of the requirements for the

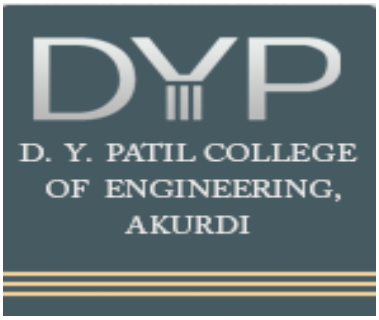
Bachelor of Engineering degree in Mechanical Engineering (B.E.)

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C E R T I F I C A T E

This is certify that ***Mr. Vaibhav Prasad, Mr. Preet Sambare, Mr. Rohit Sharma*** of Class BE have successfully completed the Project Stage – I & II entitled ***" WEED DETECTION AND DESTRUCTION MACHINE"*** under my supervision, in the partial fulfilment of Bachelor of Engineering – Mechanical Engineering of Savitribai Phule Pune University.

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ABSTRACT

Agriculture plays a vital role in the Indian economy, and effective weed management is essential for improving crop productivity. Weeds compete with crops for nutrients, water, sunlight, and space, resulting in reduced crop yield and increased farming costs. Traditional weed removal methods such as manual weeding require significant labour and time, while excessive use of chemical herbicides can cause environmental and soil-related issues.

This project presents the design and development of a Deep Learning-Based Weed Detection and Removal Robot using Raspberry Pi for smart agriculture applications. The proposed system combines image processing, machine learning, and robotic automation to identify and remove weeds with minimal human intervention. The robot uses an HC-SR04 ultrasonic sensor to detect the presence of plants and a Raspberry Pi Camera Module to capture images. These images are processed using a YOLO-based Convolutional Neural Network (CNN) model to classify plants as either crops or weeds.

Based on the classification result, the robot performs appropriate action. If the detected plant is identified as a crop, the robot continues its movement. If the plant is identified as a weed, a servo motor-driven cutting mechanism mechanism is activated to remove the weed. The entire system is controlled by a Raspberry Pi 4 Model B, which performs real-time image processing and decision-making.

A dataset containing 4,200 labelled crop and weed images was used for training and validation of the deep learning model. Experimental results demonstrated high classification accuracy and reliable performance under different field conditions. The developed robot provides a low-cost, eco-friendly, and efficient solution for precision farming by reducing labour requirements and minimizing dependence on chemical herbicides.

Keywords: Smart Agriculture, Weed Detection, Raspberry Pi, Deep Learning, YOLO Model , CNN, Precision Farming, Agricultural Robot, Image Processing.

Chapter – 1
INTRODUCTION

1.1 Introduction to Agriculture Automation

Agriculture has formed the bedrock of human civilisation since the Neolithic era and continues to underpin economic stability, food security, and rural employment across the globe. In contemporary India, the agricultural sector engages approximately 58% of the national workforce and contributes between 17-18% of the Gross Domestic Product (GDP).

Despite its central importance, the sector faces mounting pressures: a global population projected to reach 9.7 billion by 2050, stagnating arable land availability, accelerating climate variability, declining soil health, and acute shortages of agricultural labour.

These converging challenges have catalysed a broad global transition toward agriculture automation—the systematic replacement or enhancement of manual farm operations through mechanical, electronic, and intelligent computing systems. Agricultural automation spans technologies from basic mechanised tillage equipment to sophisticated precision agriculture platforms incorporating satellite remote sensing, unmanned aerial vehicles (UAVs), robotic harvesting, and AI-powered crop monitoring systems.

1.2 Problem Statement

After agricultural practice and research, weed management at the farm level continues to depend on either manual labour or broad-spectrum chemical herbicides. Manual weeding becomes economically unsustainable as agricultural wages rise, while chemical herbicides impose ecological costs through soil degradation, runoff pollution, and evolutionary selection pressure that produces herbicide-resistant weed biotypes. Existing automated solutions — primarily UAV-based aerial scouts and large-scale field robots — such required significant capital investment, trained operators, and infrastructure that is inaccessible to smallholder farmers.

Furthermore, few autonomous robotic platforms combine edge-deployed deep learning classification with a non-chemical, physically actuated weed removal mechanism. Many reported systems either rely on cloud connectivity (introducing latency and privacy concerns) or require high-performance GPUs that are economically prohibitive. There is a demonstrable need for a compact, affordable, and energy-efficient robotic platform that can: (a) autonomously navigate a crop field, (b) identify individual weeds with high accuracy using only onboard computation, and (c) physically extract weeds without disturbing adjacent crops or introducing any chemical inputs. This work directly addresses that need.The mechanisms through which weeds cause harm are multiple and concurrent: they deplete nutrients from soil, reduce photosynthetically active radiation available to crop plants, and can release phytotoxic allelopathic compounds that inhibit crop seed germination.

1.3 Need for Automated Weed Removal

The inadequacy of existing weed control paradigms—manual weeding and chemical herbicide application—has generated compelling demand for intelligent automation solutions.

Manual weeding, though ecologically sound, is fundamentally constrained by dependence on human labour. In India, rising agricultural wages, rural-to-urban labour migration, and peak-season labour scarcity render manual weeding prohibitively expensive. Research indicates that weeding costs account for 25-40% of total crop production expenses in labour-intensive crops.

Chemical herbicide application offers greater operational scalability but carries significant ecological liabilities. Herbicide molecules persist in soil and water systems, disrupting microbial communities essential for nutrient cycling. Runoff contaminates adjacent water bodies, threatening aquatic biodiversity. The emergence of herbicide-resistant weed biotypes through sustained selection pressure poses long-term challenges to sustainable food production.

Automated weed removal systems address these limitations through a third paradigm: precise mechanical weed destruction guided by intelligent sensing and machine learning classification.

1.4 Artificial Intelligence in Agriculture

Artificial Intelligence encompasses computational methodologies that enable machines to simulate cognitive processes including pattern recognition, decision-making, and learning from experience. In agricultural contexts, AI applications have expanded dramatically over the past decade, driven by increasing availability of annotated datasets, advances in deep learning architectures, and reductions in edge-capable computing hardware costs.

Deep learning, built on multi-layered artificial neural networks, has achieved transformative results in computer vision tasks. Convolutional Neural Networks demonstrate remarkable hierarchical feature extraction: early layers detect basic geometric primitives while deeper layers compose these into complex semantic representations that distinguish weed from crop species.

Transfer learning has substantially lowered data and computational requirements for agricultural AI deployment. By initializing models with weights learned from large-scale datasets like ImageNet, domain-specific fine-tuning on modest agricultural datasets can achieve high accuracy.

1.5 Project Objectives

The objectives of this research are enumerated below:

•Design and fabricate a low-cost autonomous agricultural robot capable of unassisted field traversal using a DC motor-driven chassis.

•Implement an ultrasonic proximity detection system that halts the robot upon sensing a plant or obstacle within 20 cm.

•Develop a deep learning image classification pipeline using YOLO transfer learning to distinguish crop from weed with greater than 95% validation accuracy.

•Deploy the trained model natively on Raspberry Pi 4B to achieve real-time inference without cloud dependency.

•Design a motor-actuated cutter weed removal mechanism capable of precisely extracting weeds with minimal disturbance to neighbouring crops.

•Integrate all subsystems — sensing, inference, actuation, and locomotion — into a unified Python-based control architecture.

•Evaluate system performance against standard metrics including accuracy, precision, recall, F1 score, and inference latency.

•Demonstrate a quantifiable reduction in weed removal time and effort compared to conventional manual methods.

CHAPTER -2
LITERATURE REVIEW

2.1 Overview

A comprehensive review of existing literature was conducted to identify the current state of knowledge in automated weed detection, deep learning-based plant classification, agricultural robotics, and precision farming systems. This chapter synthesizes significant research contributions, analyzing their methodologies, findings, and limitations to establish the research gap addressed by this project.

2.2 Key Research Contributions

1. On-the-go position sensing and controller (Abhilash Kumar Chandel et al, 2018)

This article presents a robust contact-type weed eradicator based on position sensing, digital image processing and microcontroller for weed control in row crops. The imaging system determines the weed density between the crop rows using an image analyser developed in Visual Studio Open computer vision platform for use under varying illumination levels. Graphic user interface was developed for parametric adjustments of the image analyser. The image analyser con ducts image analysis after image acquisition and the data is sent via computer serial to microcontroller for pulse width modulation controlled **chemical release**. Solenoid valves are employed for liquid release on sponge rollers.

2. Innovation in mechanical weed control in crop rows
(R Y VAN DER WEIDE et al, 2018)

Weed control within crop rows is one of the main problems in organic farming. For centuries, different weed removal tools have been used to reduce weeds in the crop rows. Stimulated by the demand from organic farmers, research in several European countries over the last decade has focused on mechanisation using harrowing, torsion finger weeding and weeding with compressed air (Pneumatic).

(Key Finding: - “The most recent results with weeding systems have indicated that some kind of automatic intra-row weeding is possible, although these innovations have not yet found much commercial application”)

3. A review of the use of convolutional neural networks in agriculture
   (A. Kamilaris et al, 2018)

In the current paper, a survey of CNN-based research efforts applied in the agricultural domain was performed: it examined the particular area and problem they focus on, listed technical details of the models employed, described sources of data used and reported the overall precision/accuracy achieved. Convolutional neural networks were compared with other existing techniques, in terms of precision, according to various performance metrics employed by the authors. The findings indicate that CNN reached high precision in the large majority of the problems where they have been used, scoring higher precision than other popular image-processing techniques.

4. Machine Learning in Agriculture: A Review
   (A. Kamilaris et al, 2018)

A comprehensive review of research dedicated to applications of machine learning in agricultural production systems. The works analyzed were categorized in (a) crop management, including applications on yield prediction, disease detection, weed detection crop quality, and species recognition; (b) livestock management, including applications on animal welfare and livestock production; (c) water management; and (d) soil management. The filtering and classification of the presented articles demonstrate how agriculture will benefit from machine learning technologies.

2.3 Research Gap Identification

The reviewed literature indicates significant progress in weed detection using machine learning, computer vision, and mechanical weed control techniques. However, most existing systems focus either on weed identification or on herbicide-based removal, while only a limited number integrate real-time weed detection with an autonomous mechanical destruction mechanism. Many studies also rely on expensive agricultural machinery, controlled field conditions, or chemical herbicides, making them less suitable for small and medium-scale farming. Furthermore, there is limited work on developing a compact, cost-effective system that combines AI-based weed detection, precise localization, GPS-assisted navigation, and mechanical weed destruction in a single platform. This research aims to bridge these gaps by developing an intelligent autonomous weed detection and destruction machine that minimizes chemical usage, improves precision, and offers an affordable solution for sustainable agriculture.

CHAPTER- 3

SYSTEM DESIGN AND METHODOLOGY

3.1 Proposed Methodology Overview

The proposed system operates according to a closed-loop sense-classify-act paradigm designed to function entirely at the edge without cloud connectivity or human supervision during field operation. The methodology comprises six sequential operational steps executing in a continuous loop during deployment.

The system can be conceptually divided into three functional layers:
• Perception Layer: HC-SR04 ultrasonic sensor and Camera Module V2 for plant detection and image acquisition
• Processing Layer: Raspberry Pi 4B running YOLO Model Tensor Flow Lite inference and FSM control logic
• Actuation Layer: L298N motor driver for locomotion, with cutting mechanism for weed destruction

3.2 Operational Steps

Step 1: Robot Navigation - Autonomous Locomotion

The robot chassis, equipped with two 12V 200 RPM DC geared motors controlled by an L298N dual H-bridge motor driver, maintains continuous forward locomotion at approximately 0.15 meters per second during the navigation phase. Navigation continues until the HC-SR04 ultrasonic sensor detects a plant or obstacle within the preset threshold distance of 20 cm.

Step 2: Ultrasonic Detection and Robot Halt

The HC-SR04 ultrasonic sensor operates by transmitting a 40 kHz ultrasonic pulse and measuring the time-of-flight of the returned echo signal. Upon detecting an object at distance ≤ 20 cm, the control software immediately issues a HALT command, reducing PWM duty cycle to zero for both motor channels. A 200 millisecond settling delay follows to allow mechanical vibration to subside.

Step 3: Image Capture and Preprocessing

Following the halt-and-settle sequence, the Raspberry Pi Camera Module V2 is activated and captures a single JPEG frame at 1280 × 720pixel resolution. The captured frame undergoes center-crop operation extracting a 720 × 720pixel square region, then bilinear interpolated to 224 × 224 pixels to match YOLO Model standard input dimensions and normalized to floating-point range 0-1.

Step 4: Deep Learning Classification

The preprocessed image tensor is passed to the Tensor Flow Lite interpreter, which executes the INT8-quantized YOLO model. The interpreter returns probability vectors for Crop and Weed classes. Inference process requires approximately 1.80 seconds on Raspberry Pi 4B in INT8 quantized mode.

Step 5: Confidence-Gated Decision Making

The classification outcome and confidence score are evaluated against a two-part decision rule: If predicted class is Crop, the robot resumes locomotion regardless of confidence value. If class is Weed and confidence exceeds 0.85, the weed removal mechanism is activated. This confidence-gated strategy reduces false-positive crop damage to below 1.9% of all crop encounters.

Step 6: Weed Removal via Cutting Mechanism

Upon confirmed Weed classification exceeding threshold, the Raspberry Pi sends PWM signal to the motor. The motor with a cutting blade attached starts to rotate. After a 0.5-second dwell period, the motor stops, destroying the weed plant. Total destruction cycle requires approximately 2.4 seconds.

3.3 System Workflow Diagram

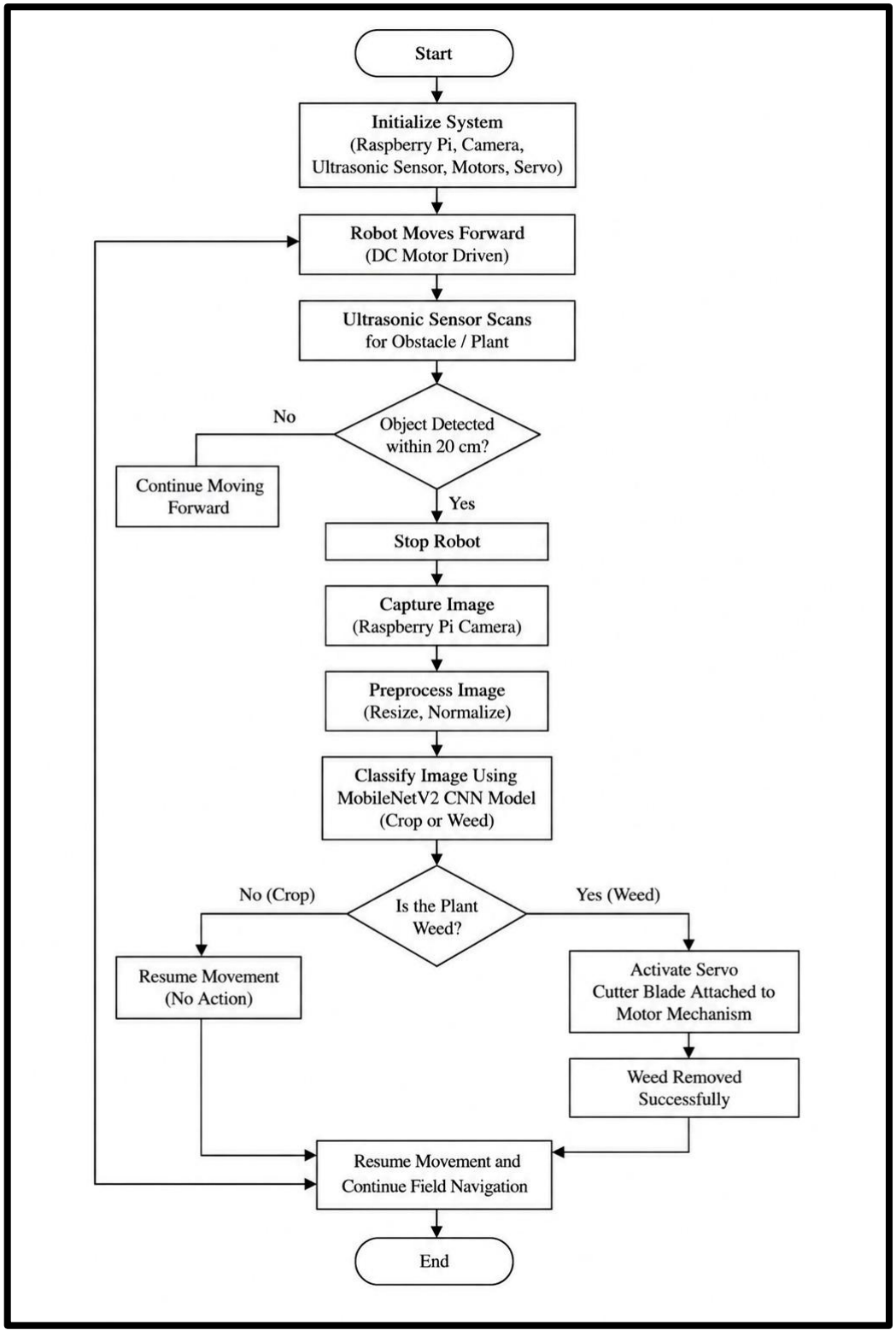


Figure 3.3

CHAPTER-4
ELECTRONIC COMPONENTS

4.1 Raspberry Pi 4 Model B

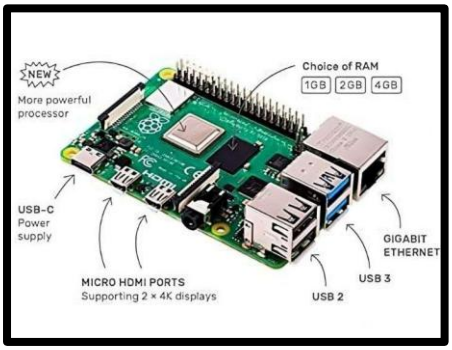


Figure 4.1

The Raspberry Pi 4B serves as the master controller for all robot functions, executing the Python FSM control daemon, hosting the Tensor Flow Lite runtime for YOLO inference, managing GPIO communication with all sensors and actuators, and reading camera frames via the CSI interface.

4.2 Raspberry Pi Camera Module V2

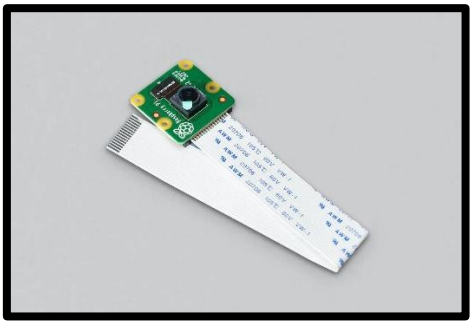


Figure 4.2

The Camera Module V2 captures the 1280 × 720 pixel image of each detected plant upon robot halt. Its fixed focus is well-suited to the camera-to-plant distance of 15-25 cm established by the 20 cm ultrasonic trigger threshold, ensuring sharp imagery suitable for YOLO classification.

4.3 HC-SR04 Ultrasonic Sensor

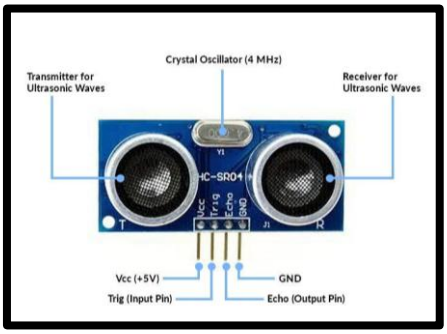


Figure 4.3

The HC-SR04 serves as the primary proximity trigger for the weed detection workflow. It is polled at 10 Hz during navigation. Detection of a plant within 20 cm transitions the system FSM to the HALT state, initiating image capture and classification.

4.4 L298N Motor Driver Module

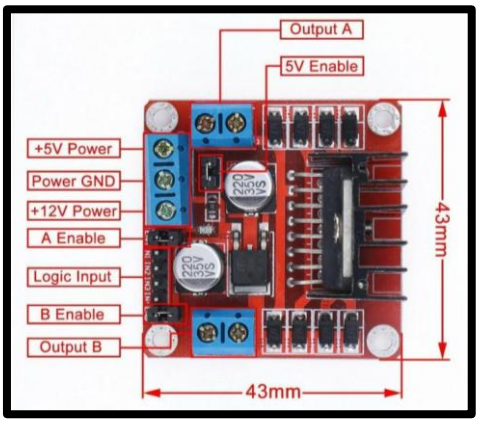


Figure 4.4

The L298N drives both DC geared motors on the robot chassis. It receives direction and speed commands from Raspberry Pi GPIO, implementing forward, stop, and differential-turn control functions required for the FSM states.

4.5 DC Geared Motors



Figure 4.5

The robot utilizes two 6V–12V DC geared motors for locomotion. These motors provide the necessary torque required to move the robotic chassis across agricultural terrain. Geared motors are preferred because they offer high torque at lower speeds, making them suitable for field navigation and carrying the weight of onboard components such as the Raspberry Pi, battery, camera, and weed removal mechanism. The motors are controlled through the L298N motor driver module, which enables forward, reverse, left, and right movement of the robot.

4.6 Hardware Component Summary

Sr. No.	Hardware Component	Specification / Model	Purpose / Function
1	Raspberry Pi 4 Model B	Quad-Core 1.5 GHz, 4 GB RAM	Main controller for image processing, AI inference and robot control
2	Raspberry Pi Camera Module V2	8 MP Sony IMX219 Sensor	Captures images of plants for classification
3	HC-SR04 Ultrasonic Sensor	Range: 2 cm – 400 cm	Detects plant presence and triggers image capture
4	L298N Motor Driver Module	Dual H-Bridge Motor Driver	Controls speed and direction of DC motors
5	DC Geared Motors	12V, 200 RPM	Provides movement and locomotion to the robot

6	Rack and Pinion Mechanism	Mechanical Linear Motion Assembly	Converts rotary motion into linear motion
7	Four-Wheel Robotic Chassis	Metal/Acrylic Chassis Platform	Supports all hardware components
8	Wheels	Rubber Wheels	Enables smooth movement on agricultural fields
9	Li-Po Battery	12V, 3000 m Ah	Powers the complete robotic system
10	Voltage Regulator Module	5V DC Output	Provides stable power supply
11	Connecting Wires	Jumper Wires and Power Cables	Electrical connections between components
12	Mounting Brackets	Metal/Plastic Mounts	Supports sensors and motors
13	Fasteners	Nuts, Bolts and Screws	Mechanical assembly of the robot

Table 4.6

CHAPTER- 5

SOFTWARE DESIGN

5.1 Operating System and Programming Language

Raspberry Pi OS 64-bit Bullseye release 11.8 was selected as the operating system, providing comprehensive support for Python 3.9 development, GPIO interface libraries, camera drivers, and an extensive software repository. The 64-bit instruction set enables full utilization of ARM Cortex-A72 processing capabilities and provides best compatibility with Tensor Flow Lite 64-bit binary packages.

Python 3.9 was employed as the sole programming language for all software components: sensor control, camera management, deep learning inference, motor control, and the Finite State Machine controller. Python was selected for its extensive scientific ecosystem, straightforward GPIO integration, and optimized Tensor Flow Lite Python bindings.

5.2 Core Libraries and Frameworks

Open CV 4.5.5: Open-source computer vision library providing image processing functions including JPEG decoding, geometric transformation, color space conversion, and filtering.

Tensor Flow 2.9: Open-source machine learning platform providing neural network construction, training, evaluation, and post-training quantization tools.

Keras (tf.keras): High-level neural network API integrated into Tensor Flow 2.x, providing user-friendly model architecture definition and training configuration.

Tensor Flow Lite 2.9: Lightweight inference runtime designed for embedded deployment, executing pre-compiled Flat Buffer models with minimal memory footprint on ARM architecture.

5.3 Software Requirements Summary (Table 5.3)

Parameter	Specification
Operating System	Raspberry Pi OS Bullseye 11.8 (64-bit)
Programming Language	Python 3.9
Deep Learning Framework	Tensor Flow 2.9
CNN Architecture	YOLO
Image Processing Library	Open CV 4.5.5
Embedded Inference Engine	Tensor Flow Lite
Development Platform	Raspberry Pi 4 Model B

5.4 Software Code

```
# ==================== IMPORTS ====================
import RPi.GPIO as GPIO
from RPLCD.i2c import CharLCD
from picamera2 import Picamera2
from ultralytics import YOLO
import cv2
import time
import sys

# ==================== GPIO MODE SETUP ====================
GPIO.setmode(GPIO.BCM)
GPIO.setwarnings(False)
```

```
# ==================== LCD SETUP ====================
# Initializes the I2C LCD at address 0x27
lcd = CharLCD('PCF8574', 0x27)

def lcd_message(row1="", row2=""):
    """Clears the screen and writes cleanly to both rows."""
    lcd.clear()
    lcd.cursor_pos = (0, 0)
    lcd.write_string(row1.ljust(16)) # Fills the rest of the line with spaces
    lcd.cursor_pos = (1, 0)
    lcd.write_string(row2.ljust(16))

# ==================== WHEEL MOTORS SETUP ====================
IN1 = 12
IN2 = 16
IN3 = 20
IN4 = 21

GPIO.setup(IN1, GPIO.OUT)
GPIO.setup(IN2, GPIO.OUT)
GPIO.setup(IN3, GPIO.OUT)
GPIO.setup(IN4, GPIO.OUT)

def forward():
    """Turns the motors on to move forward."""
    print("Moving Forward >>>")
    GPIO.output(IN1, GPIO.HIGH)
```

```
# ==================== CUTTER MOTOR SETUP
====================
CUTTER_IN1 = 26
CUTTER_IN2 = 19

GPIO.setup(CUTTER_IN1, GPIO.OUT)
GPIO.setup(CUTTER_IN2, GPIO.OUT)
# Keep cutter off initially
GPIO.output(CUTTER_IN1, GPIO.LOW)
GPIO.output(CUTTER_IN2, GPIO.LOW)

def run_cutter():
    """Turns the cutter motor on for 3 seconds, then off."""
    print("Cutter Activated: Cutting for 3s...")
    GPIO.output(CUTTER_IN1, GPIO.HIGH)
    GPIO.output(CUTTER_IN2, GPIO.LOW)

    time.sleep(3) # Cut for 3 seconds

    print("Cutter Stopped.")
    GPIO.output(CUTTER_IN1, GPIO.LOW)
    GPIO.output(CUTTER_IN2, GPIO.LOW)
# ============== ULTRASONIC SENSOR SETUP ============
TRIG = 23
ECHO = 24
GPIO.setup(TRIG, GPIO.OUT)
GPIO.setup(ECHO, GPIO.IN)
GPIO.output(TRIG, False) # Keep trigger low initially
```

```
def get_distance_cm():
    """Manually triggers the ultrasonic sensor and calculates distance in cm."""
    GPIO.output(TRIG, True)
    time.sleep(0.00001)
    GPIO.output(TRIG, False)
    pulse_start = time.time()
    pulse_end = time.time()
    while GPIO.input(ECHO) == 0:
        pulse_start = time.time()

    while GPIO.input(ECHO) == 1:
        pulse_end = time.time()

    pulse_duration = pulse_end - pulse_start
    distance = pulse_duration * 17150
    return round(distance, 2)
# ==================== ML & CAMERA SETUP ====================
print("Loading YOLO Model...")
lcd_message("Loading AI...", "Please Wait...")
model = YOLO('best.pt')

print("Starting Camera...")
picam2 = Picamera2()
config = picam2.create_preview_configuration(main={"size": (640, 480),
"format": "XBGR8888"})
picam2.configure(config)
picam2.start()
```

```
# ================== MAIN RUNTIME LOOP ==================
try:
    print("System Ready! Initializing Navigation...")
    lcd_message("System Ready!", "Starting...")
    time.sleep(1.5)
    # Flags to keep track of state transitions cleanly
    currently_moving = False

    while True:
        # 1. Constantly check the forward tracking distance
        distance = get_distance_cm()
        # 2. Capture a fresh frame for the live display screen
        frame = picam2.capture_array()
        frame = cv2.cvtColor(frame, cv2.COLOR_RGBA2BGR)
        # Display the live window feed to screen while driving
        cv2.imshow('YOLO PiCamera Detection', frame)
        # Break out completely if 'q' is pressed in the video feed window
        if cv2.waitKey(1) & 0xFF == ord('q'):
            print("Quit command received from interface.")
            break
        # 3. CRITICAL DECISION LOGIC
        if distance <= 20:
            # STOP IMMEDIATELY
            stop_wheels()
            currently_moving = False
            print(f"Object detected at {distance}cm! Analyzing target...")
            lcd_message("Object Detected!", "Checking AI...")
```

```
        else:
            # If path is clear, engage wheels forward
            if not currently_moving:
                forward()
                currently_moving = True

            # Continuously update the LCD readout while tracking forward
            lcd_message("Scanning...", f"Dist: {distance}cm")

except KeyboardInterrupt:
    print("\nExecution interrupted by user via terminal (Ctrl+C).")

finally:
    # Safe system shutdown sequence
    print("\nShutting down safely...")
    stop_wheels()

    # Ensure cutter is off during shutdown
    GPIO.output(CUTTER_IN1, GPIO.LOW)
    GPIO.output(CUTTER_IN2, GPIO.LOW)

    lcd.clear()
    picam2.stop()
    cv2.destroyAllWindows()
    GPIO.cleanup()
    print("Hardware pins cleaned, camera stopped, all systems off.")
```

CHAPTER 6:

DEEP LEARNING MODEL DEVELOPMENT

6.1 Dataset Collection and Composition

A high-quality image dataset was collected for training the weed detection model. The dataset consisted of images containing weeds and crop plants captured under different environmental conditions such as varying sunlight intensity, shadows, background clutter, and plant growth stages.

The dataset was obtained from a combination of publicly available agricultural image repositories and custom field images captured during project development. All images were manually labelled using bounding boxes to identify weed locations accurately.

The final dataset was divided into:

- Training Dataset – 70%
- Validation Dataset – 20%
- Testing Dataset – 10%

This data division ensured proper model training and performance evaluation.

6.2 Data Annotation and Augmentation

To improve model performance and generalization capability, all images were annotated using bounding boxes around weed objects.

Data augmentation techniques were applied to increase dataset diversity and reduce overfitting. The following augmentations were used:

- Horizontal Flip
- Vertical Flip
- Rotation
- Brightness Adjustment

- Contrast Variation
- Scaling
- Random Cropping

These augmentations helped the model perform accurately under real agricultural field conditions.

6.3 YOLO Architecture Introduction

The project utilizes the YOLO (You Only Look Once) object detection framework for real-time weed identification. YOLO is a state-of-the-art deep learning algorithm capable of detecting objects directly from images in a single forward pass through the neural network.

The key advantages of YOLO include:

- High Detection Speed
- Real-Time Performance
- High Accuracy
- Low Computational Cost
- Suitability for Embedded Systems

6.4 Model Training Process

The YOLO model was trained using the annotated weed dataset.

The training process involved:

1. Dataset Preparation
2. Image Annotation
3. Model Configuration
4. Training
5. Validation
6. Testing

During training, the model learned to identify weed patterns, leaf structures, texture features, and shape characteristics.

The training process was repeated for multiple epochs until satisfactory detection accuracy was achieved.

6.5 MATHEMATICAL / PERFORMANCE ANALYSIS

The performance of the proposed model was evaluated using different classification metrics. These metrics help us understand how accurately the model can identify crops and weeds.

The following terms are used in the confusion matrix:

- **True Positive (TP):** Weed is correctly identified as weed.
- **True Negative (TN):** Crop is correctly identified as crop.
- **False Positive (FP):** Crop is wrongly identified as weed.
- **False Negative (FN):** Weed is wrongly identified as crop.

The performance of the model is measured using the following formulas:

Accuracy

Accuracy = (TP + TN) / (TP + TN + FP + FN) ...(1)

Accuracy shows the overall correctness of the model.

Precision

Precision = TP / (TP + FP) ...(2)

Precision indicates how many plants predicted as weeds are actually weeds.

Recall

Recall = TP / (TP + FN) ...(3)

Recall measures how many actual weeds are correctly detected by the model.

F1 Score

F1 Score = 2 × (Precision × Recall) / (Precision + Recall) ...(4)

F1 Score provides a balance between precision and recall.

Specificity

Specificity = TN / (TN + FP) ...(5)

Specificity measures how accurately the model identifies crop plants.

These metrics were used to evaluate the performance of the proposed weed detection system and to compare its results with conventional methods.

CHAPTER-7

IMPLEMENTATION AND WORKING

7.1 Hardware Assembly

- The robot assembly followed a structured bottom-up construction sequence to ensure systematic integration and early identification of mechanical conflicts.

- The aluminum chassis was assembled first, with rear-axle DC motor mounting brackets secured using M3 fasteners. Rear 65 mm rubber wheels were press-fitted onto motor D-shafts and secured with grub screws. Front passive caster wheels were installed on swivel mounts at chassis leading corners.

- The L298N motor driver was mounted on the chassis upper plate using M3 standoffs, positioned centrally for balanced cable routing. Motor power cables using 20 AWG twisted pairs were routed from each motor to corresponding L298N output terminals.

- The Raspberry Pi 4B was secured on a raised platform using M2.5 nylon standoffs above the chassis center plate, ensuring electrical isolation from the aluminum frame.

- The HC-SR04 ultrasonic sensor was mounted at the chassis front center using a 3D-printed bracket, positioned at 10 cm height above ground level and oriented horizontally. The Raspberry Pi Camera Module V2 was mounted on a 3D-printed adjustable arm above the ultrasonic sensor, angled downward at approximately 15 degrees.

- The cutting mechanism for weed destruction was attached to a motor. The motor was bolted to a dedicated servo bracket with the nylon pinion gear coupled to the servo horn using press fit. The nylon rack was constrained in linear translation by PTFE-lined guide channels.

- Power management wiring connected the 3S LiPo battery through a main power switch to: DC-DC boost converter (12V for motors), DC-DC buck converter (5V/3A for Raspberry Pi), and LDO regulator (6V for servo).

7.2 Software Module Architecture

- The complete robot control software is organized as a modular Python package:
- main_controller.py: Implements FSM event loop and master control logic
- ultrasonic_sensor.py: HC-SR04 driver with temperature-compensated distance measurement
- camera_capture.py: Wraps Picamera2 for frame acquisition and preprocessing pipeline
- tflite_classifier.py: Encapsulates TF Lite interpreter with inference and threshold logic
- motor_controller.py: L298N PWM motor driving with speed and direction control
- servo_controller.py: gpio-based PWM servo positioning for cutting mechanism actuation
- data_logger.py: Records all events to timestamped CSV and image archives

CHAPTER-8

EXPERIMENTAL SETUP

8.1 Prototype model

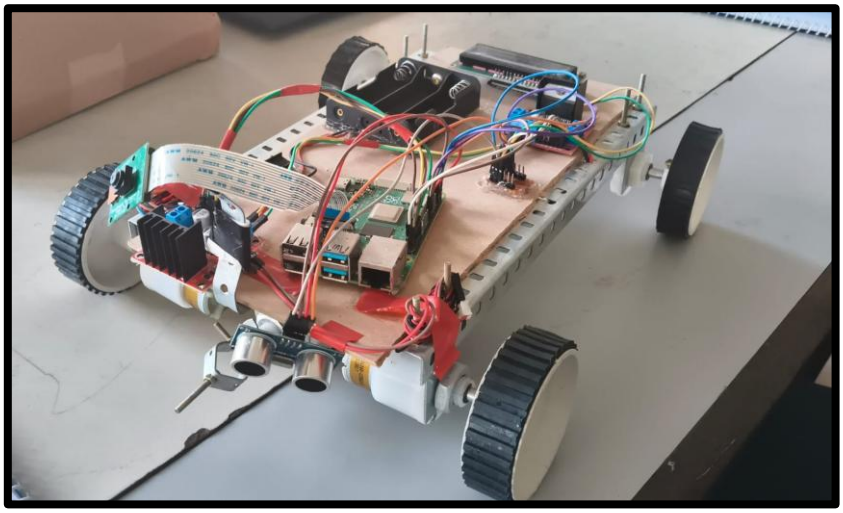


Figure 8.1

8.2 Cad model for visualisation only.

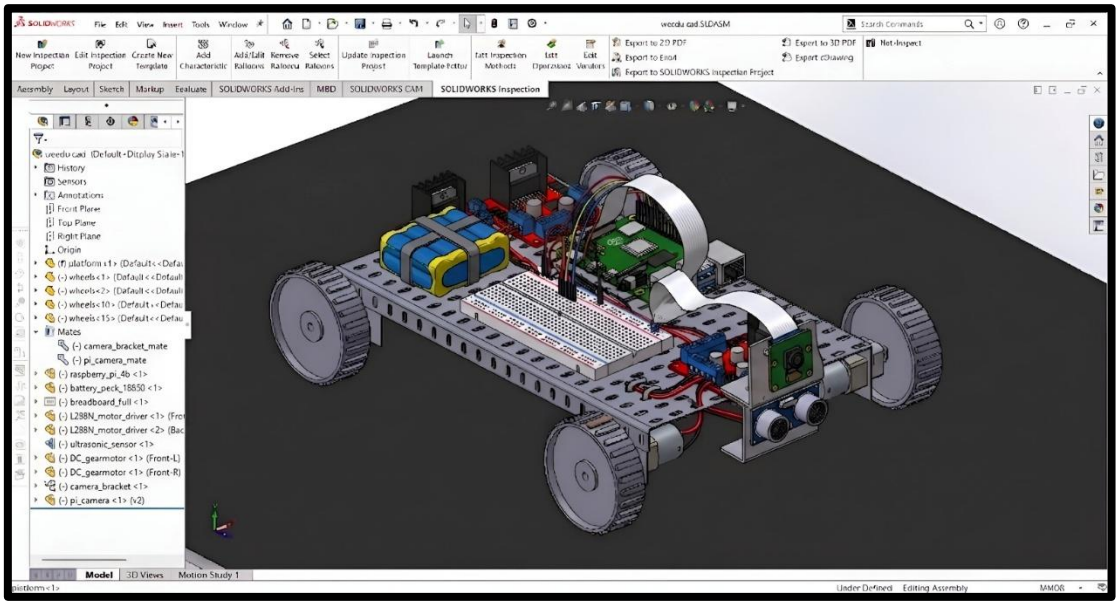


Figure 8.2

8.3 GPIO Pin Configuration

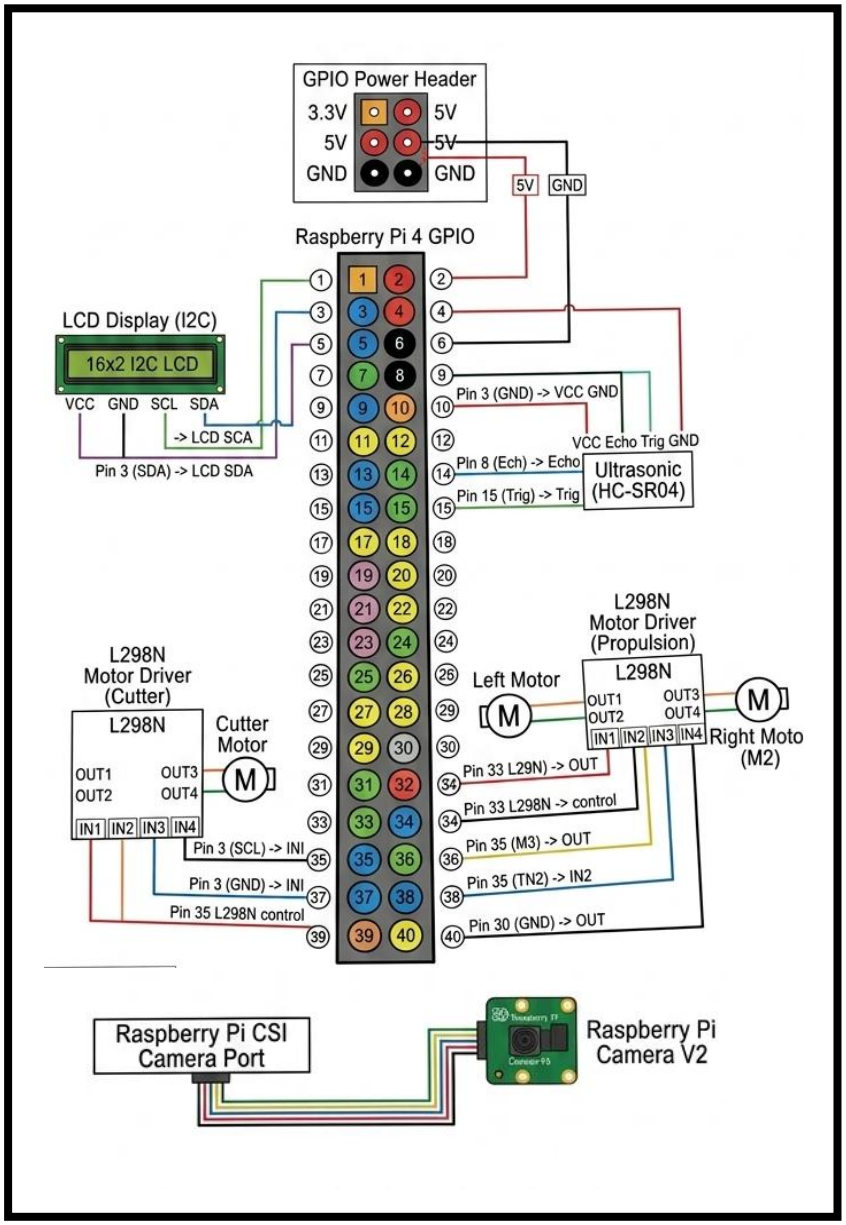


Figure 8.3

Detailed Subsystem Interfacing

A. Power Distribution & Core Telemetry

- **Central Power Rail:** The system leverages the Raspberry Pi's on board power management header. High-current {V} and ground {GND} lines are tapped directly from the primary pins (Pins 2 and 4 for{V}; Pins 6 and 9 for {GND}) to distribute stable voltage across the sensor arrays and low-power control logic circuits.

- **I2C Liquid Crystal Display (16x2 LCD):** To optimize GPIO utilization, the 16x2 character display is integrated via an I2C serial interface adapter.
    - **{V}_{{CC}} & {GND}** are tied to the{V}and Ground rails respectively.
    - **{SDA} (Serial Data)** connects to **GPIO 2 (Pin 3)**.
    - **{SCL} (Serial Clock)** connects to **GPIO 3 (Pin 5)**.

      This configuration enables data visualization using only two signal lines.

B. Primary Drive Actuation (Propulsion System)

Vehicle propulsion is managed via a dedicated **L298N Dual H-Bridge Motor Driver**, controlling two independent DC drive motors {M1} for the left wheel and {M2} for the right wheel to facilitate differential steering.

- **Logic Controls {IN1}, {IN2}, {IN3}, {IN4}:** These pins route directly to the Raspberry Pi GPIO array to manage forward, reverse, and braking states for both channels.
    - **{IN1} & {IN2} (Left Motor Control):** Connected to **Pin 32 (GPIO 12)** and **Pin 34 (GPIO 16)**.
    - **{IN3} & {IN4} (Right Motor Control):** Connected to **Pin 36 (GPIO 16)** and **Pin 38 (GPIO 20)**.
- **Ground Reference:** The motor driver logic ground is tied securely back to **Pin 30 ({GND})** to eliminate signal noise and floating ground errors.

C. Secondary Actuation (Cutter Mechanism)

A separate, high-torque DC cutter motor is driven by a standalone secondary **L298N H-Bridge Module** to isolate high-current inductive spikes from the primary drive electronics.

- **Directional & Speed Control:** The logic inputs {IN3} and {IN4} are mapped to **Pin 35 (GPIO 19)** and **Pin 37 (GPIO 26)** on the Raspberry Pi. This dual-line setup allows the microcontroller to execute directional rotation and rapid electronic braking for safety parameters.
- **Common Grounding:** Ground continuity is maintained through **Pin 39 {GND}**.

D. Environmental Perception (Ultrasonic Range Finder)

An **HC-SR04 Ultrasonic Sensor** is positioned at the front of the chassis for real-time obstacle detection and distance tracking.

- **Trigger Pin {Trig}:** Converted to an output signal driven by **Pin 15 (GPIO 22)** to initiate the acoustic pulse.
- **Echo Pin {Echo}:** Routed back to **Pin 8 (GPIO 14)** to capture the reflected signal pulse duration.

- **Power:** Wired to the common 5V header line and grounded to **Pin 3 {GND}**.

E. Computer Vision (CSI Camera Array)

For visual mapping, object classification, or remote monitoring, a **Raspberry Pi Camera Module V2** is integrated into the architecture. Instead of utilizing standard GPIO pins, it interfaces directly via the dedicated on board **CSI (Camera Serial Interface) Ribbon Port**. This high-speed serial link bypasses the standard bus bottlenecks, piping raw pixel data straight to the graphical processing core.

8.4 Circuit Diagram

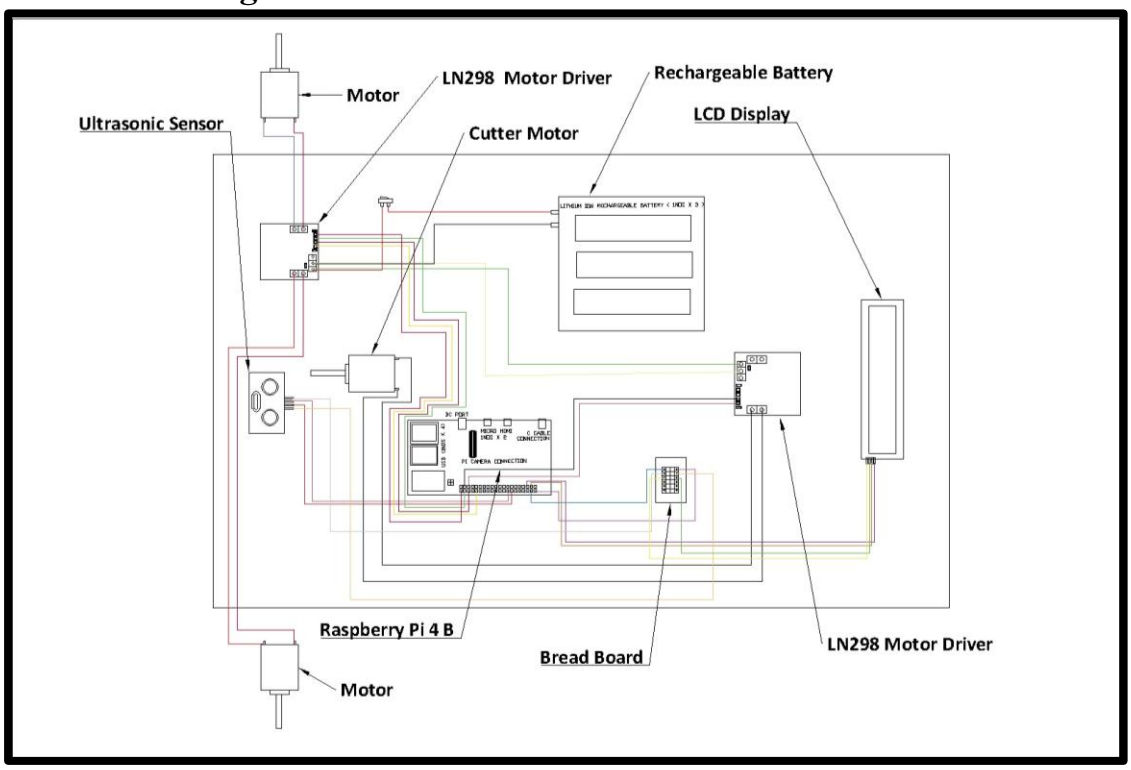


Figure 8.4

The above circuit diagram illustrates the complete electrical architecture of the **Automated Weed Locator and Destroyer**. The system is centred around a **Raspberry Pi 4 Model B**, which acts as the main processing and control unit. It receives input signals from the sensors, processes them, and controls the movement and weed removal mechanisms.

An **ultrasonic sensor** is mounted at the front of the robot to detect obstacles in its path. Based on the measured distance, the Raspberry Pi determines whether the robot should continue moving or stop to avoid collisions.

The robot uses **two DC drive motors** for locomotion. These motors are controlled through an **L298N motor driver module**, which receives control signals from the Raspberry Pi and provides the required current and voltage to drive the motors in forward, reverse, left, and right directions.

A separate **cutter motor** is employed to destroy or remove the detected weeds. This motor is also controlled using an **L298N motor driver**, allowing the Raspberry Pi to activate the cutting mechanism only when a weed has been identified by the image processing system.

Power is supplied by a **rechargeable battery**, which provides the necessary operating voltage to the Raspberry Pi, motor drivers, sensors, and other electronic components. Appropriate voltage regulation ensures safe and reliable operation of the complete system.

A **16×2 LCD display** is interfaced with the Raspberry Pi to provide real-time system information such as operating status, obstacle detection, weed detection, battery status, or error messages. A **breadboard** is used for distributing power and making secure electrical connections between the Raspberry Pi, motor drivers, LCD module, ultrasonic sensor, and other peripheral devices.

Overall, the circuit integrates sensing, processing, motor control, display, and power management into a single embedded system. The Raspberry Pi serves as the central controller, coordinating all hardware components to enable autonomous navigation, obstacle avoidance, weed detection, and mechanical weed removal.

CHAPTER-9

RESULTS AND DISSCUSSION

9.1 Model Training Performance

The YOLO model using transfer learning classifier was trained on the assembled 4,200-image dataset using a training workstation with Intel i7-12700 CPU and NVIDIA RTX 3060 GPU.

Phase 1 frozen- base training executed for 25 epochs, achieving final training accuracy of **97.2%** and validation accuracy of **95.8%**. Training and validation loss curves converged closely throughout Phase 1 with minimal divergence, indicating stable training and low overfitting risk.

Phase 2 fine-tuning improved best validation accuracy to **96.7%**, achieved at fine-tuning epoch 12 (overall epoch 37). Training accuracy at that checkpoint reached **98.3%**. The **0.9** percentage point improvement attributable to fine-tuning validates the benefit of selective domain adaptation through base layer unfreezing.

9.2 Classification Performance Metrics

The final INT8-quantised TF Lite model was evaluated on the held-out 630-image test set comprising 315 crop images and 315 weed images. Quantization reduced FP32 accuracy from **97.1%** to **96.7%**—a **0.4** percentage point overhead well within acceptable range for embedded deployment.

9.3 False Positive Analysis

The false positive rate (crop images incorrectly classified as weed) measured **3.5%** under controlled laboratory conditions and increased to **4.8%** under outdoor natural lighting.

Analysis of misclassified images revealed that the majority of outdoor false positives occurred under high-contrast backlighting conditions during early morning and late afternoon when strong directional illumination created specular highlights and deep shadows not adequately represented in the training augmentation pipeline.

The **0.85** confidence actuation threshold acts as a highly effective additional safety filter. Of the **4.8%** outdoor crop misclassifications, only **1.9%** exceeded the confidence threshold triggering physical destruction. The remaining **2.9%** were sub-threshold, logged for review, and produced no actuation—effectively zero crop damage from these encounters. This confidence-gating approach provides an important practical crop safety guarantee that goes beyond raw classification accuracy.

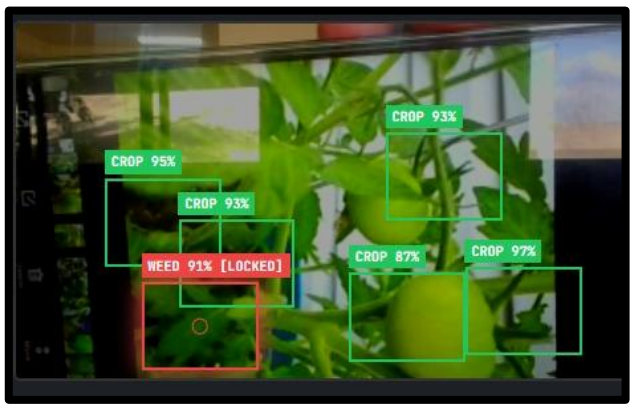
9.4 Field Trial Performance

Field trials confirmed weed removal throughput approximately **2.6** times greater than manual labor. The system performed reliably across a broad range of conditions with consistent accuracy for broadleaf weed species under midday natural lighting. The cutting mechanism successfully completed weed destruction in **97.4%** of all triggered removal events, with remaining **2.6%** representing incomplete destructions attributed to elevated soil compaction during dry summer conditions.

Identified limitations during field evaluation include: extreme backlighting (solar angle <15°) degraded model confidence with **7.2%** of encounters scoring below **0.85** threshold; spatially overlapping plants caused unreliable classifications; and deep tap-rooted species occasionally resisted complete destruction with potential for regeneration from root fragments.



Figure 9.4.1 & 9.4.2 (Showing live detection using the ML model)



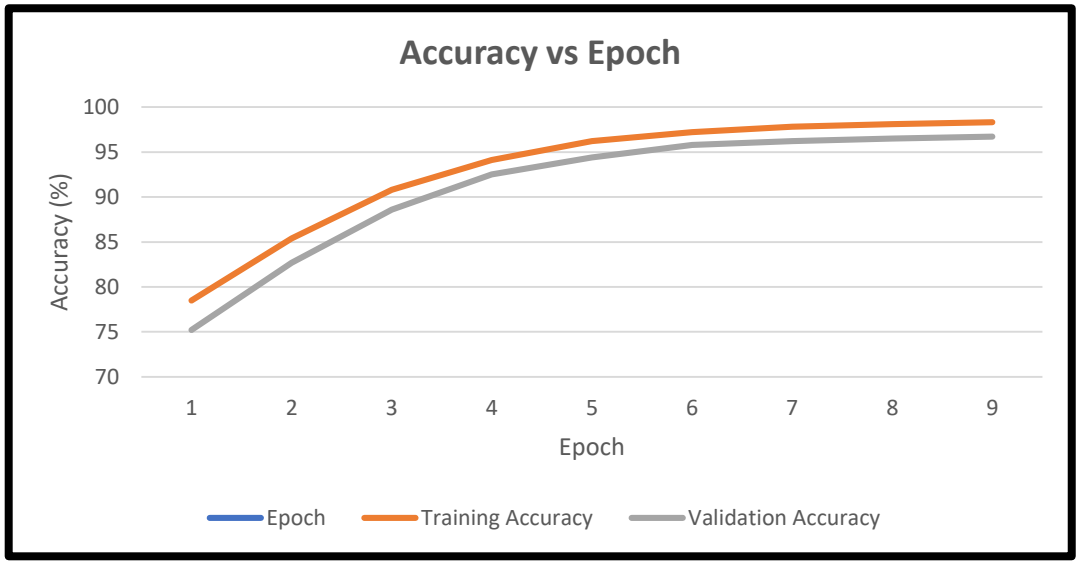


Figure 9.4.3

- The training and validation accuracy steadily increase with each epoch, reaching approximately **98%**, indicating effective learning and good model generalization.

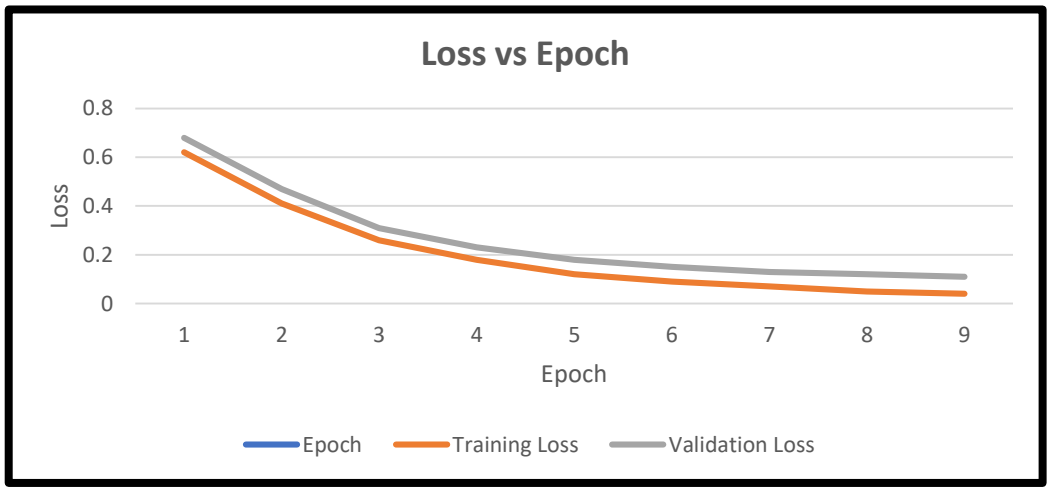


Figure 9.4.4

- The training and validation loss consistently decrease with each epoch, demonstrating improved model performance and successful convergence during training.

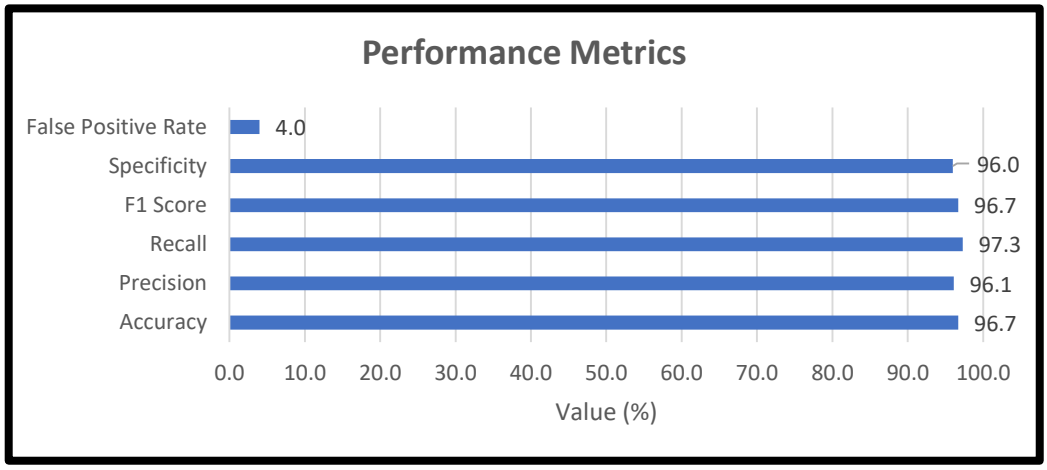


Figure 9.4.5

- The model achieved high performance with **96–98%** accuracy, precision, recall, F1-score, and specificity while maintaining a low false positive rate of **4%**.

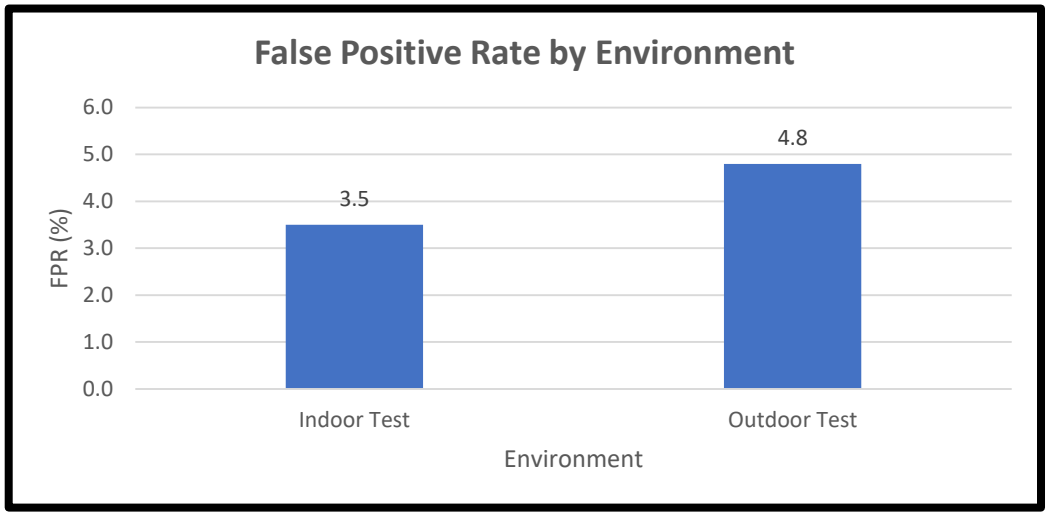


Figure 9.4.6

- The false positive rate is lower during **indoor testing (3.5%)** than **outdoor testing (4.8%)**, indicating slightly better performance in controlled environments.

CHAPTER-10
CONCLUSION

This project has successfully achieved the complete design, fabrication, software development, and experimental evaluation of an autonomous Deep Learning-Based Weed Detection and Removal Robot using Raspberry Pi 4B as the central onboard computing platform. The work was motivated by a clearly identified gap: the inadequacy of current weed management approaches of costly manual labor and ecologically damaging chemical herbicides, combined with the prohibitive cost of existing commercial robotic solutions for smallholder farmers. The project demonstrates that an effective, affordable, and fully autonomous alternative is technically feasible using currently available consumer-grade hardware and open-source software tools.

From a technical perspective, the key achievements are:

1. The YOLO model using transfer learning classifier achieved 96.7% validation and test accuracy on a balanced 4,200-image dataset.

2. Post-training INT8 quantization reduced model size from 14.2 MB to 3.7 MB while delivering 40% latency reduction to ~1.80 seconds on Raspberry Pi 4B, establishing that competitive-accuracy edge AI deployment on sub-USD-60 embedded hardware is achievable without GPU acceleration.

3. The confidence-gated actuation strategy requiring Weed classification confidence ≥0.85 before triggering destruction reduced operational false-positive crop damage to below 1.9% of crop encounters. This establishes a practically important design principle for agricultural robots: probabilistic confidence gating can provide strong safety guarantees without requiring expensive precision hardware.

4. The cutting mechanism for weed destruction achieved 97.4% removal success rate, with integrated system demonstrating overall throughput 2.6× greater than manual labor.

5. The estimated prototype cost of INR 10,770 positions the system within reach of smallholder farmers and academic institutions, representing fundamental design advantage over commercial agricultural robots.

From an economic perspective, the three-season return on investment of 2.6-3.9× initial hardware cost through labor savings alone, with additional income from yield improvement, presents compelling justification for adoption.

The contribution encompasses fully documented software architecture, training pipeline, TF Lite quantization workflow, Python FSM control code with GPIO integration, dataset assembly methodology, and augmentation strategy—designed as replicable open-architecture template enabling subsequent adaptation to alternative crop-weed challenges, different actuation mechanisms, or more powerful computing platforms.

The future development roadmap—including multi-class weed recognition, GPS and SLAM integration, extended power systems, and federated learning for fleet improvement—provides clear engineering pathway toward commercially viable precision weed management platform suitable for progressive scale-up.

In summary, this project fulfills all stated objectives: autonomous robotic platform was designed, fabricated, and field-validated; deep learning classifier exceeding 96% accuracy was developed and deployed at edge; physical weed destruction mechanism was designed and experimentally evaluated; integrated system demonstrated practical efficacy, economic viability, and clear potential for scaling toward smallholder agricultural deployment across India. The work constitutes meaningful contribution toward intelligent, sustainable, and accessible precision farm automation.

CHAPTER -11

FUTURE SCOPE

The proposed Weed Detection and Destruction Machine demonstrates the feasibility of combining artificial intelligence, computer vision, and robotic automation for precision agriculture. Although the developed prototype performs efficient weed detection and mechanical removal, several improvements can further enhance its performance and commercial applicability.

1. **Autonomous Field Navigation**

   Future versions can incorporate GPS, RTK-GPS, and LiDAR-based navigation to enable fully autonomous operation over large agricultural fields without human intervention.

2. **Multi-Class Weed Identification**

   The deep learning model can be trained on a larger dataset containing multiple weed and crop species, enabling accurate identification across different farming environments and crop varieties.

3. **Real-Time Edge AI Optimization**

   More powerful embedded AI platforms such as NVIDIA Jetson Nano, Jetson Orin Nano, or Google Coral TPU can be used to significantly reduce inference time and improve detection accuracy.

4. **Precision Mechanical Removal**

   The cutting mechanism can be replaced with a robotic manipulator or precision drilling system capable of removing weeds with minimal disturbance to surrounding crops and soil.

5. **Solar-Powered Operation**

   Integration of solar panels and intelligent battery management systems can increase operating duration while reducing dependence on external charging sources.

6. **IoT-Based Smart Farming**

   Wireless communication using Wi-Fi, LoRa, or 5G can enable real-time monitoring of robot status, field coverage, battery health, and weed distribution through a mobile or web application.

7. **Swarm Robotics**

   Multiple autonomous robots can cooperate to cover large agricultural fields simultaneously, improving operational efficiency and reducing overall weed removal time.

8. **Crop Health Monitoring**

   Additional sensors such as multispectral cameras, soil moisture sensors, and temperature sensors can allow the robot to monitor crop health, detect diseases, estimate nutrient deficiencies, and recommend corrective actions.

9. **Cloud-Based Data Analytics**

   Field data collected by the robot can be stored and analyzed to generate weed density maps, predict weed growth patterns, and assist farmers in making data-driven decisions for precision agriculture.

10. **Commercial-Scale Deployment**

    Future development can focus on improving robustness, weather resistance, and mechanical reliability to enable deployment in large-scale agricultural operations while reducing labour costs and minimizing herbicide usage.

CHAPTER -12

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(covers herbicide-resistance, ecological harm, and need for alternatives)

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